to compile it. For my 64-bit set-up, I compiled and installed *libopenotp* with *make && make install*. I then moved */usr/local/lib/libopenotp** to */usr/lib64*.

After getting *libopenotp* into the appropriate directory, run an *ldconfig*.

Next, compile *pam_otp* with *make*. Copy *pam_openotp.so* to the architecture-specific *security* directory: for 64-bit, */lib64/security/* and for 32-bit, */lib/security*. On 64-bit, remove the *pam* module from */lib/security/*.

Edit */etc/pam.d/sshd* and replace the line with *"auth include system-auth"* with the following, replacing *myserver* with the WebADM server's IP address:

```
auth required pam_openotp.so server_url="http://myserver:8080/
openotp/" client_id="SSH"
```

My file now looks like this:

```
auth    required    pam_env.so
auth   required pam_openotp.so server_
url=http://192.168.0.1:8888/openotp/ client_id="SSH" password_
mode=2, default_domain="example"
```

Note: Read the *README* file in your OpenOTP PAM module distribution for configuration details.

Next, let's configure the OpenSSH service. Edit */etc/ssh/sshd_config*. Enable OpenSSH challenge response mode with "*ChallengeResponseAuthentication yes*". Enable OpenSSH password authentication with "*PasswordAuthentication yes*". Enable OpenSSH with PAM with "*UsePAM yes*". Save and exit. Restart the SSH service.

Now if you try to connect as the test user (for example, *ssh testuser@192.168.0.1)* from any box, it will prompt you for an OTP password, and not the normal password.

RADIUS Bridge (RB)

OpenOTP RADIUS Bridge provides the RADIUS API for OpenOTP. It is an optional server component to be deployed on your OpenOTP installation and is implemented over the open source FreeRADIUS software.

Installation and configuration

Download RADIUS Bridge from *http://www.rcdevs.com/downloads/* and install it using the self-installer script (for example, *sh radiusd.1.0.6-1.sh*). The installation creates the RADIUS Bridge system user and sets filesystem permissions. Edit */opt/radiusd/conf/openotp.conf* and change the following parameters, adapting them to your environment:

```
Server_url = http://192.168.0.1:8080/openotp/
Password_mode = 2
Default_domain = "example"
Data_is_vps = yes
```

For more information regarding the settings, please look at the *README* file.

Start the radius service with */opt/radiusd/bin/radius restart*.

If required, add your RADIUS network access clients (for example, your VPN server IP address) in */opt/radiusd/conf/clients.conf*. Each client must be configured with an IP address and a RADIUS secret. By default, RADIUS Bridge accepts requests from any client using the RADIUS secret "testing123".

If you want the RADIUS server to reply with a value-attribute pair, make sure all the attribute and value pairs exist in the */opt/radiusd/lib/dictionaries/dictionary.XXX* file, and specify the same in the "*Reply-Data*" section of the user (in the WebADM GUI console) as follows:

```
192.168.0.1:Vendor-XXX-Profile-ID=2000,192.168.0.2:Vendor-XXX-
Profile-ID=3000
```

This means that if "testuser" logs in from 192.68.0.1, the network access client will get a value of 2000, while if the person logs in from 192.168.0.2. it will get a value of 3000.

Note: The RADIUS Attribute Value Pairs (AVP) carry data in both the request and the response for the authentication, authorisation, and accounting transactions.

We are done!

OpenOTP is pretty vast, and everything can't be covered in a single article. We covered the basic set-up and configuration. For other services, please check the documentation available at the website.

This product is scalable and you can also configure the services in HA mode. For more information please check the product website.

In case of problems, I am also very reachable by e-mail! Contact me at *neomatrixgem@gmail.com*.

If you want to go ahead with two-factor authentication, and don't want to invest, then you can also check OPIE, Google Authenticator, etc. END

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